

ECE 461 - HW 10

1.a) $g(t) = f(t) \cos 10\pi t$

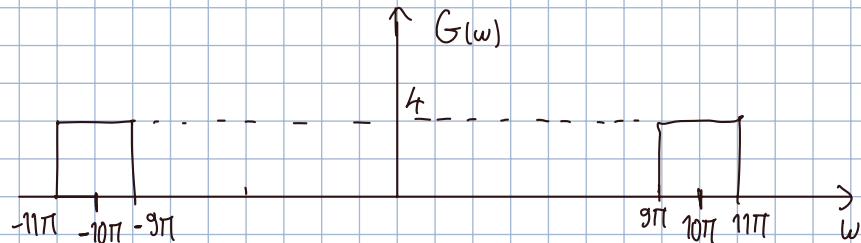
Taking FT of both sides (we will work with ω formalism not f):

$$G(\omega) = \frac{1}{2\pi} [F(\omega) * \mathcal{FT}\{\cos 10\pi t\}]$$

convolution

$$\mathcal{FT}\{\cos 10\pi t\} = \int_{-\infty}^{\infty} \frac{e^{j10\pi t} + e^{-j10\pi t}}{2} e^{-j\omega t} dt = \pi [\delta(\omega - 10\pi) + \delta(\omega + 10\pi)]$$

$$G(\omega) = \frac{1}{2} F(\omega - 10\pi) + \frac{1}{2} F(\omega + 10\pi)$$

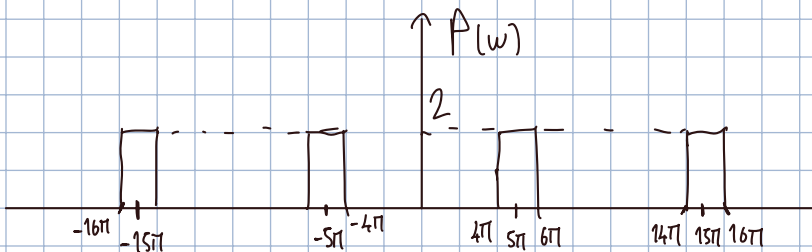


1.b) $p(t) = f(t) \cos(10\pi t) \cos(5\pi t) = f(t) \frac{\cos(15\pi t) + \cos(5\pi t)}{2}$

Similarly $\mathcal{FT}\{\cos(5\pi t)\} = \pi [\delta(\omega - 5\pi) + \delta(\omega + 5\pi)]$

$$P(\omega) = G(\omega) * \left[\frac{1}{2} \delta(\omega - 5\pi) + \frac{1}{2} \delta(\omega + 5\pi) \right]$$

$$P(\omega) = \frac{1}{4} [F(\omega - 15\pi) + F(\omega + 15\pi) + F(\omega - 5\pi) + F(\omega + 5\pi)]$$



* Note that this is equivalent to sum of $F(\omega)$ with shifts 5π and 15π as also apparent from the time domain form of $p(t)$, scaled with $1/4$

1.c) $q(t) = f(t) \cos(10\pi t) \cos^2(5\pi t) = \frac{f(t)}{2} [\cos(15\pi t) \cos(5\pi t) + \cos(5\pi t) \cos(5\pi t)]$

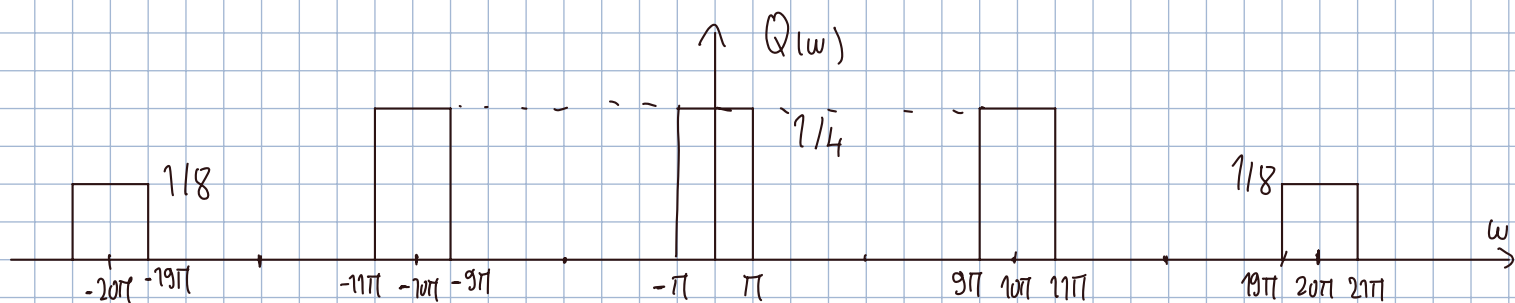
$$= \frac{f(t)}{4} [\cos(20\pi t) + \cos(10\pi t) + \cos(10\pi t) + 1]$$

$$= \frac{f(t)}{4} [1 + 2\cos(10\pi t) + \cos(20\pi t)]$$

$$Q(\omega) = P(\omega) * \left[\frac{1}{2} \delta(\omega - 5\pi) + \frac{1}{2} \delta(\omega + 5\pi) \right]$$

$$= \frac{1}{8} \left[F(\omega - 20\pi) + F(\omega - 10\pi) + F(\omega - 10\pi) + F(\omega) + F(\omega) + F(\omega + 10\pi) + F(\omega + 10\pi) + F(\omega + 20\pi) \right]$$

$$= \frac{1}{8} \left[F(\omega - 20\pi) + F(\omega + 20\pi) + 2(F(\omega - 10\pi) + F(\omega + 10\pi)) + 2F(\omega) \right]$$

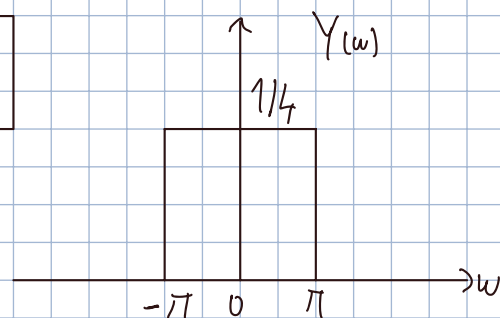


1.d) We will first consider freq. domain now.

Signal $q(t)$ passes through filter $H(\omega)$, meaning $Y(\omega) = Q(\omega)H(\omega)$

Since $H(\omega)$ is a low-pass filter with gain 1 b/w $|\omega| < \pi$ and it completely rejects frequencies with $|\omega| > 2\pi$, filter will only pick the central rectangle $1/4 F(\omega)$ in $Q(\omega)$. Thus;

$$Y(\omega) = \frac{1}{4} F(\omega)$$



→ Taking the inverse Fourier Transform gives

$$y(t) = \frac{1}{4} f(t)$$

$$2) \quad x(t) = \text{sinc}(t) \cos(2\pi f_0 t) + 0. j \sin(2\pi f_0 t)$$

$$h(t) = \text{sinc}^2(t) (\cos 2\pi f_0 t - \sin 2\pi f_0 t)$$

2.a)

$$x(t) = \text{Re} \{ x_+(t) \} = \text{Re} \{ \underbrace{\text{sinc}(t)}_{x_e(t): \text{bandpass equivalent}} e^{j2\pi f_0 t} \}$$

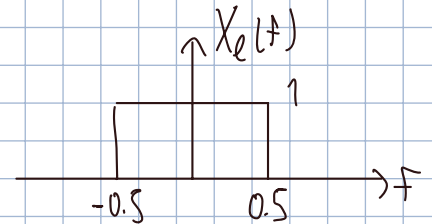
$$x_+(t) = \text{sinc}(t) e^{j2\pi f_0 t}$$

$$x_e(t) = \text{sinc}(t)$$

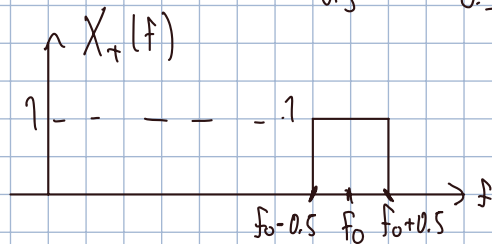
$$x_c(t) = \text{sinc}(t) \Rightarrow \{ \text{Re} \{ x_e(t) \} \}$$

$$x_s(t) = 0 \Rightarrow \text{no imaginary component}$$

$$x_e(t) = \mathcal{FT} \{ \text{sinc}(t) \} = \text{rect}(f) = \begin{cases} 1 & |f| < 0.5 \\ 0 & \text{o.w.} \end{cases}$$

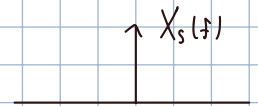


$$x_+(f) = \mathcal{FT} \{ \text{sinc}(t) \} \Big|_{f: f-f_0} = \text{rect}(f-f_0)$$

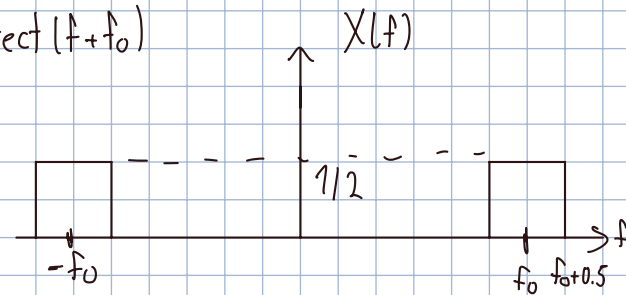


$$x_c(f) = \text{rect}(f) \quad \text{Graph of } X_c(f) \text{ is a rectangular pulse from } -0.5 \text{ to } 0.5 \text{ with height } 1.$$

$$x_s(f) = 0$$

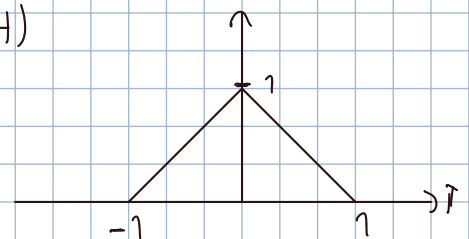


$$X(f) = \frac{1}{2} [X_+(f) + X_+^*(-f)] = \frac{1}{2} \text{rect}(f-f_0) + \frac{1}{2} \text{rect}(f+f_0)$$



2.b) Let us first calculate the FT of $\text{sinc}^2(t)$

$$\text{sinc}^2(t) \xrightarrow{\mathcal{F}} \underbrace{\text{rect}(f) * \text{rect}(f)}_{\text{convolve 2 rects}} = \underbrace{\Delta(f)}_{\text{triangular pulse}}$$



$$x(t) = A(t) \cos(2\pi f_0 t + \theta(t)) = \underbrace{A(t) \cos(\theta(t))}_{X_c(t)} \cos 2\pi f_0 t - \underbrace{A(t) \sin(\theta(t))}_{X_s(t)} \sin 2\pi f_0 t$$

Thus for $h(t) = \underbrace{\text{sinc}^2(t)}_{h_c(t)} \cos 2\pi f_0 t - \underbrace{\text{sinc}^2(t)}_{h_s(t)} \sin 2\pi f_0 t$

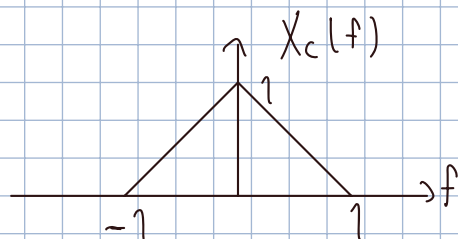
$$h_c(t) = \text{sinc}^2(t)$$

$$h_s(t) = \text{sinc}^2(t)$$

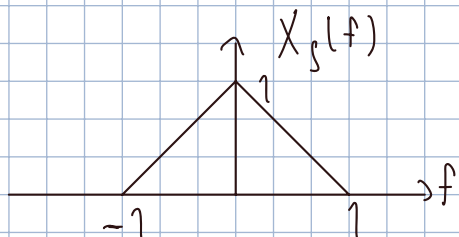
$$h_e(t) = x_c(t) + j x_s(t) = (1+j) \text{sinc}^2(t)$$

$$h_+(t) = x_e(t) e^{j2\pi f_0 t} = (1+j) \text{sinc}^2(t) e^{j2\pi f_0 t}$$

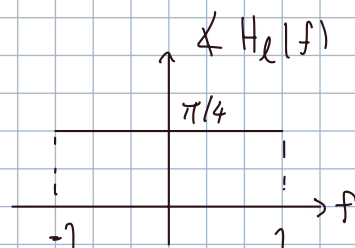
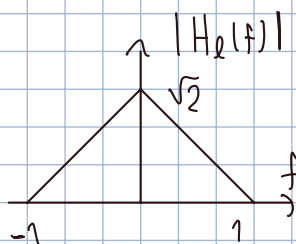
• $H_c(f) = \Delta(f) = \begin{cases} 1+f & , -1 < f < 0 \\ 1-f & , 0 \leq f < 1 \\ 0 & , \text{o.w.} \end{cases}$



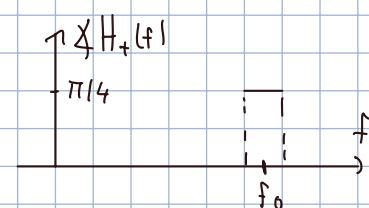
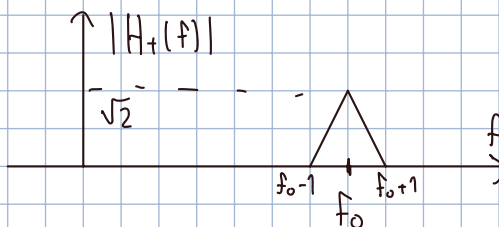
• $H_s(f) = \Delta(f)$



• $H_e(f) = (1+j) \Delta(f) = e^{j\pi/4} \sqrt{2} \Delta(f)$



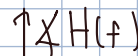
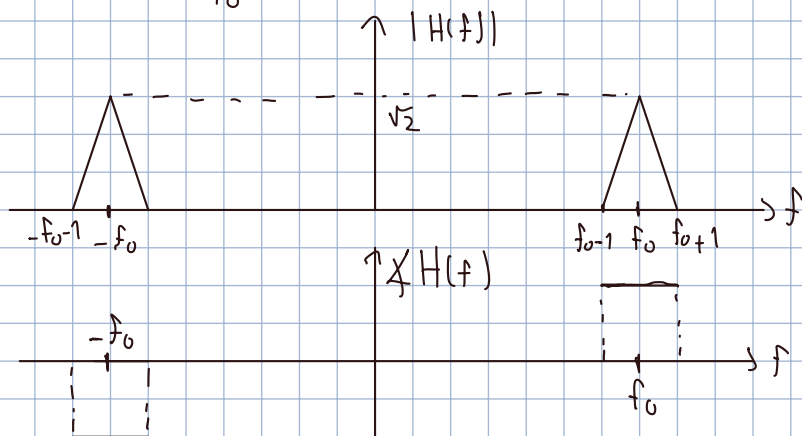
• $H_+(f) = e^{j\pi/4} \sqrt{2} \Delta(f-f_0)$



• $H(f) = \frac{1}{2} [H_+(f) + H_+^*(f)]$

$$= \frac{1+j}{2} \Delta(f-f_0) + \frac{1-j}{2} \Delta(f+f_0)$$

$$= e^{j\pi/4} \sqrt{2} \Delta(f-f_0) + e^{-j\pi/4} \sqrt{2} \Delta(f+f_0)$$



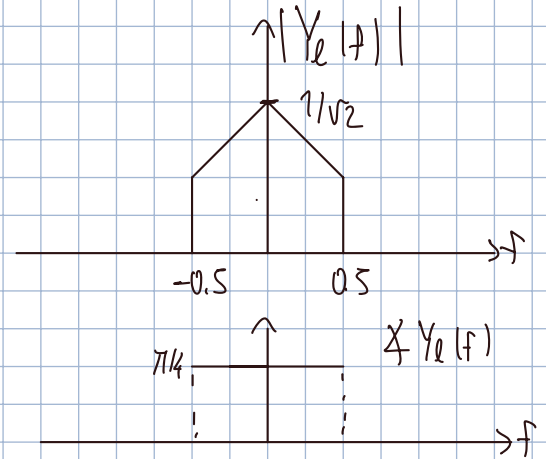
2.c) For bandpass system : $Y_L(t) = \frac{1}{2} [X_L(t) * h_L(t)]$

which translates to $Y_L(f) = \frac{1}{2} X_L(f) H_L(f)$ in freq. domain.

$$Y_L(f) = \frac{1}{2} [\text{rect}(f) \Delta(f)]$$

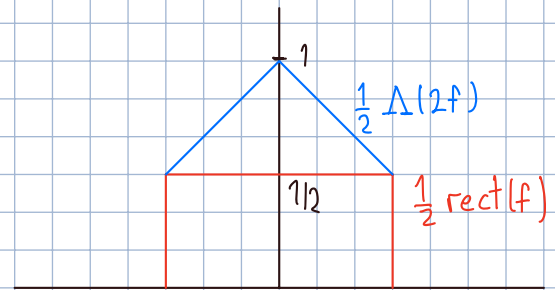
$$= \frac{1+j}{2} \begin{cases} 1-|f|, & |f| < 0.5 \\ 0, & \text{o.w.} \end{cases}$$

$$\frac{1}{\sqrt{2}} e^{j\pi/4}$$



$$Y_L(t) = \frac{1+j}{2} \mathcal{F}^{-1} \{ \text{rect}(f) \Delta(f) \}$$

* Now notice that $\text{rect}(f) \Delta(f)$ is actually sum of a $\text{rect}(f)$ with amplitude $1/2$ and a triangular pulse with width 1 ($\Delta(2f)$)

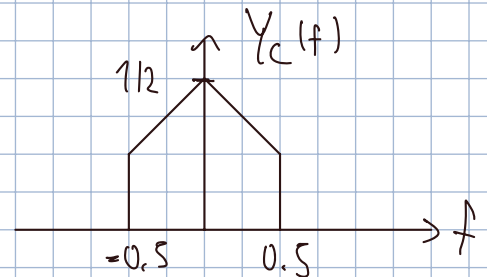


$$\begin{aligned} \mathcal{F}^{-1} \{ \text{rect}(f) \Delta(f) \} &= \mathcal{F}^{-1} \left\{ \frac{1}{2} \text{rect}(f) \right\} + \mathcal{F}^{-1} \left\{ \frac{1}{2} \Delta(2f) \right\} \\ &= \frac{1}{2} \text{sinc}(t) + \frac{1}{4} \text{sinc}^2(t/2) \end{aligned}$$

$$Y_L(t) = \frac{1+j}{2} \left[\frac{1}{2} \text{sinc}(t) + \frac{1}{4} \text{sinc}^2(t/2) \right]$$

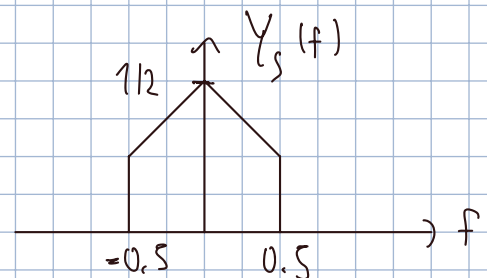
$$Y_C(t) = \text{Re} \{ Y_L(t) \} = \frac{1}{4} \text{sinc}(t) + \frac{1}{8} \text{sinc}^2(t)$$

$$Y_C(f) = \frac{1}{2} \text{rect}(f) \Delta(f) = \begin{cases} \frac{1}{2} (1-|f|), & |f| < 0.5 \\ 0, & \text{o.w.} \end{cases}$$



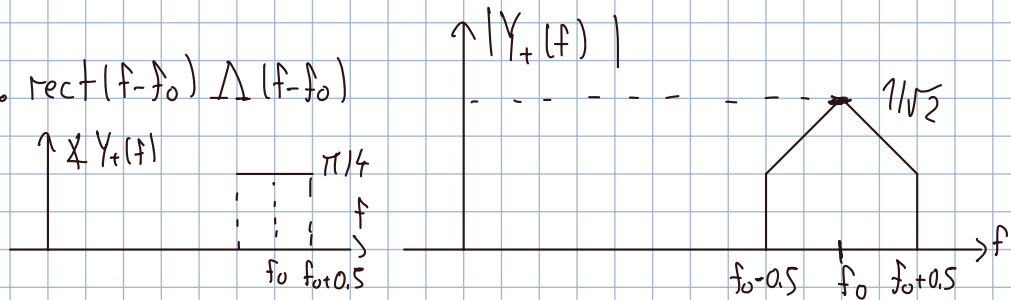
$$Y_S(t) = \text{Im} \{ Y_L(t) \} = \frac{1}{4} \text{sinc}(t) + \frac{1}{8} \text{sinc}^2(t)$$

$$Y_S(f) = \frac{1}{2} \text{rect}(f) \Delta(f) = \begin{cases} \frac{1}{2} (1-|f|), & |f| < 0.5 \\ 0, & \text{o.w.} \end{cases}$$



$$\bullet \quad Y_+(t) = Y_L(t) e^{j 2\pi f_0 t} = \frac{1+j}{2} \left[\frac{1}{2} \text{sinc}(t) + \frac{1}{4} \text{sinc}^2(t) \right] e^{j 2\pi f_0 t}$$

$$Y_+(f) = Y_L(f-f_0) = \frac{e^{j\pi/4}}{\sqrt{2}} \cdot \text{rect}(f-f_0) \Delta(f-f_0)$$



$$\bullet \quad Y(f) = \frac{1}{2} [Y_+(f) + Y_+^*(-f)] = \frac{1+j}{4} \cdot \text{rect}(f-f_0) \Delta(f-f_0) + \frac{1-j}{4} \cdot \text{rect}(f+f_0) \Delta(f+f_0)$$

